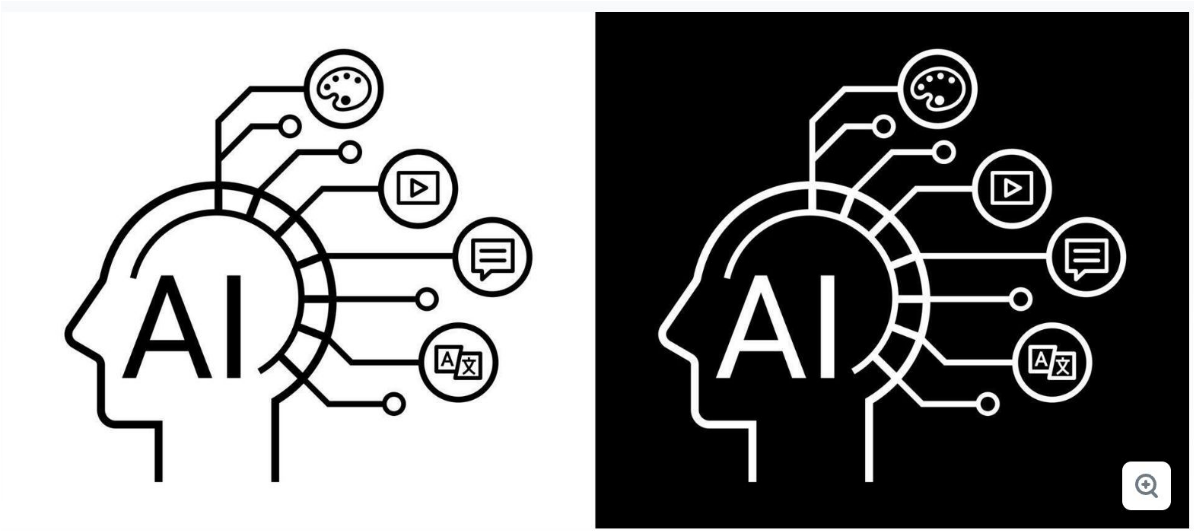
AIMS 5740
Generative Artificial Intelligence

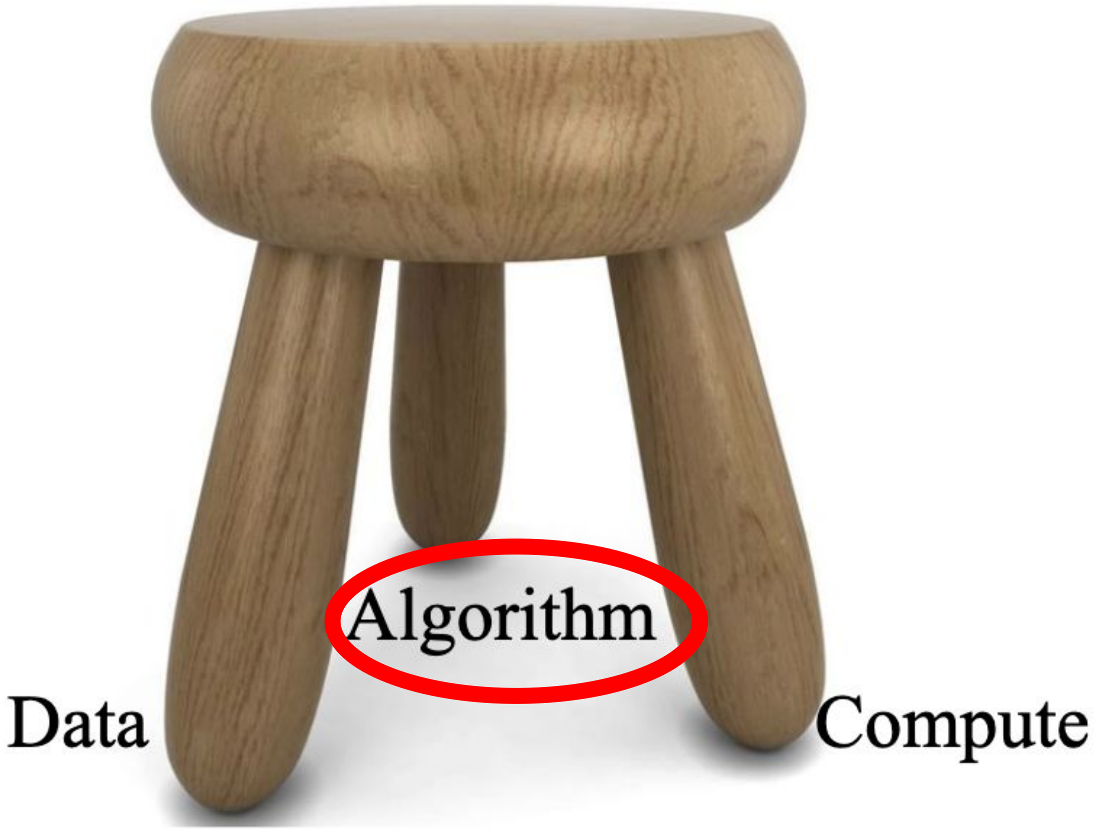
(2026 Term 2)



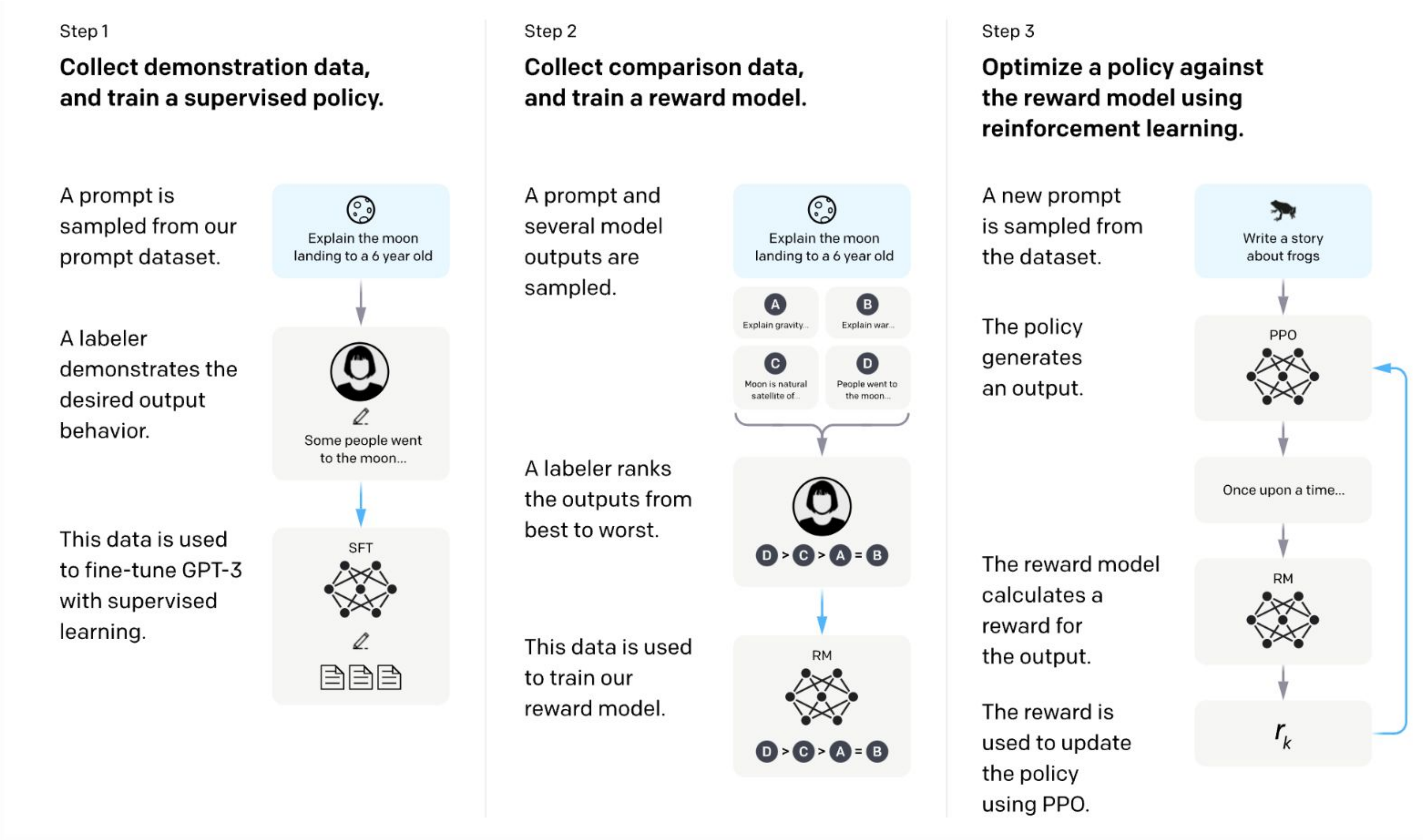
Computer Science & Engineering
The Chinese University of Hong Kong

Generative AI

- Post-training algorithms.



RLHF


Step 1
Collect demonstration data, and train a supervised policy.
A prompt is sampled from our prompt dataset.
Explain the moon landing to a 6 year old
A labeler demonstrates the desired output behavior.
Some people went to the moon...
This data is used to fine-tune GPT-3 with supervised learning.
SFT
Step 2
Collect comparison data, and train a reward model.
A prompt and several model outputs are sampled.
Explain the moon landing to a 6 year old
A Explain gravity...
B Explain war...
C Moon is natural satellite of...
D People went to the moon...
A labeler ranks the outputs from best to worst.
D > C > A = B
This data is used to train our reward model.
RM
D > C > A = B
Step 3
Optimize a policy against the reward model using reinforcement learning.
A new prompt is sampled from the dataset.
Write a story about frogs
The policy generates an output.
PPO
Once upon a time...
The reward model calculates a reward for the output.
RM
The reward is used to update the policy using PPO.
r_k

Human Feedback

- Human reward

SAN FRANCISCO, California (CNN) -- A magnitude 4.2 earthquake shook the San Francisco ... overturn unstable objects.

An earthquake hit San Francisco. There was minor property damage, but no injuries.

s_1

$$R(s_1) = 8.0$$

The Bay Area has good weather but is prone to earthquakes and wildfires.

s_2

$$R(s_2) = 1.2$$

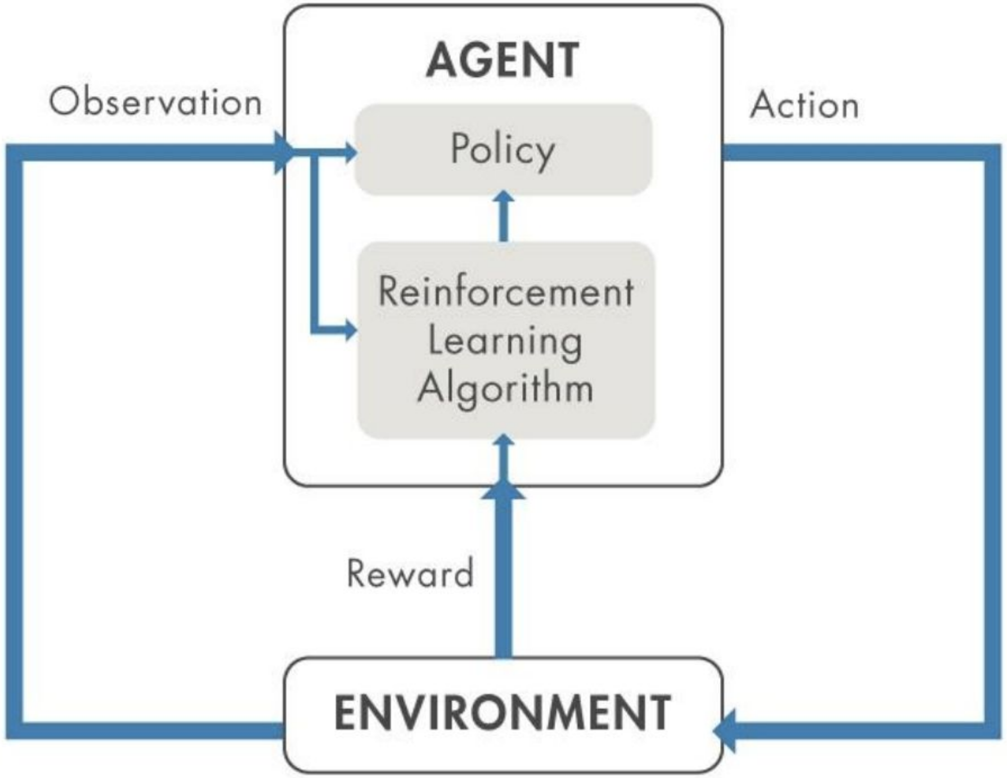
Goal: maximize the expected reward of samples from our LM

$$\mathbb{E}_{\hat{s} \sim p_\theta(s)}[R(\hat{s})]$$

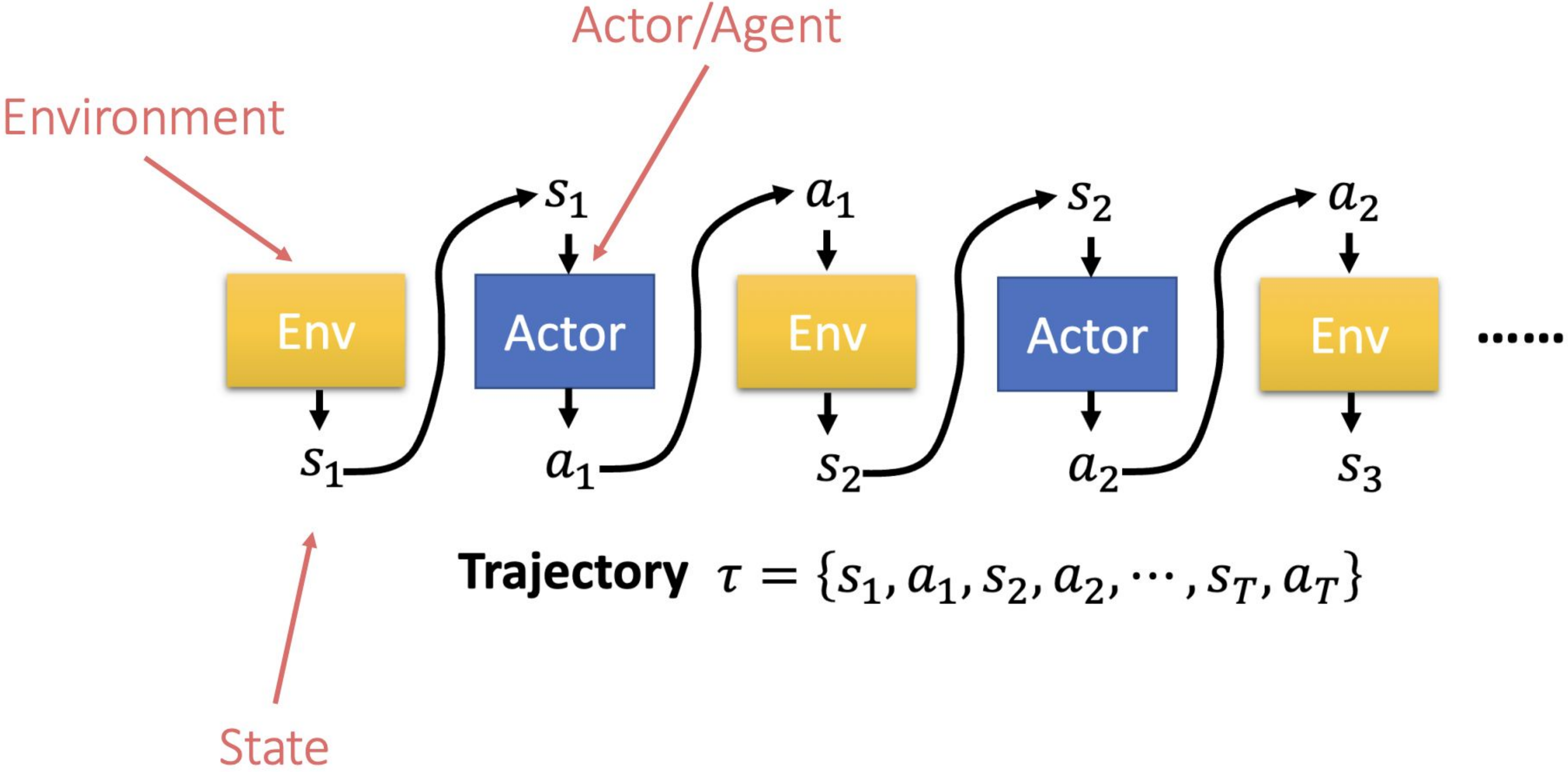
Reinforcement Learning

How do we change the LM parameters θ to maximize this?

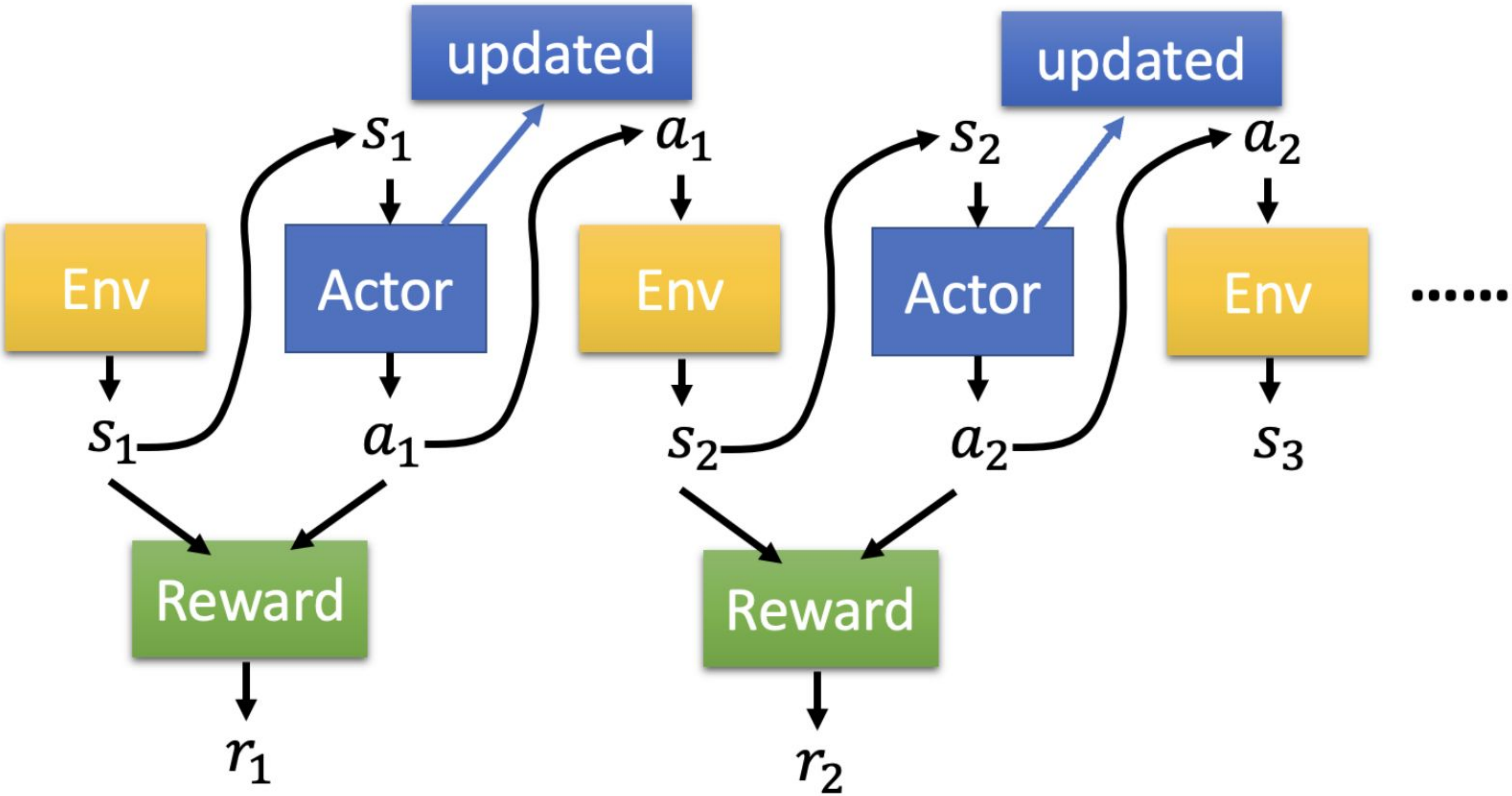
$$\mathbb{E}_{\hat{s}\sim p_\theta(s)}[R(\hat{s})]$$



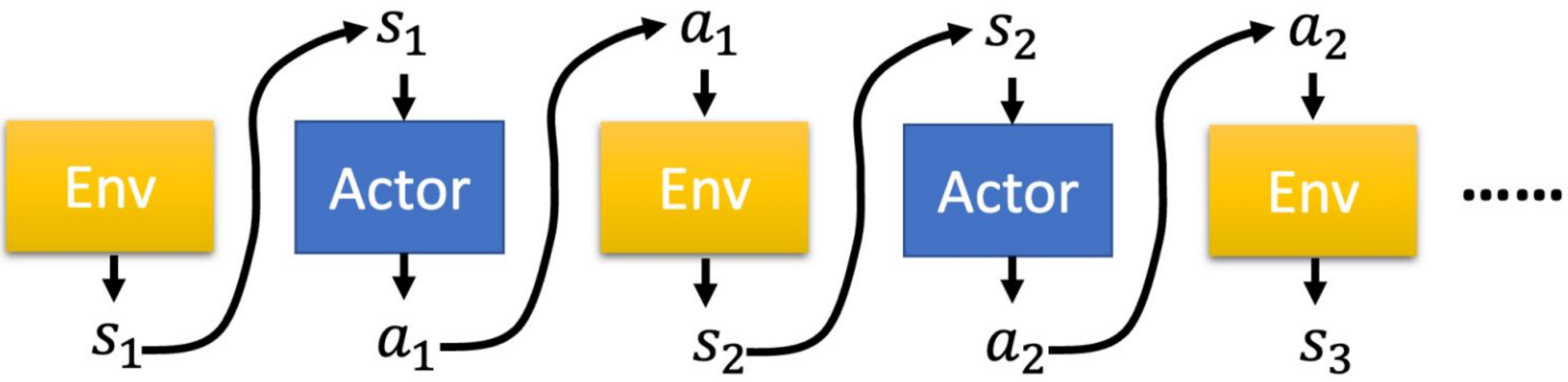
Reinforcement Learning



Reinforcement Learning



Reinforcement Learning

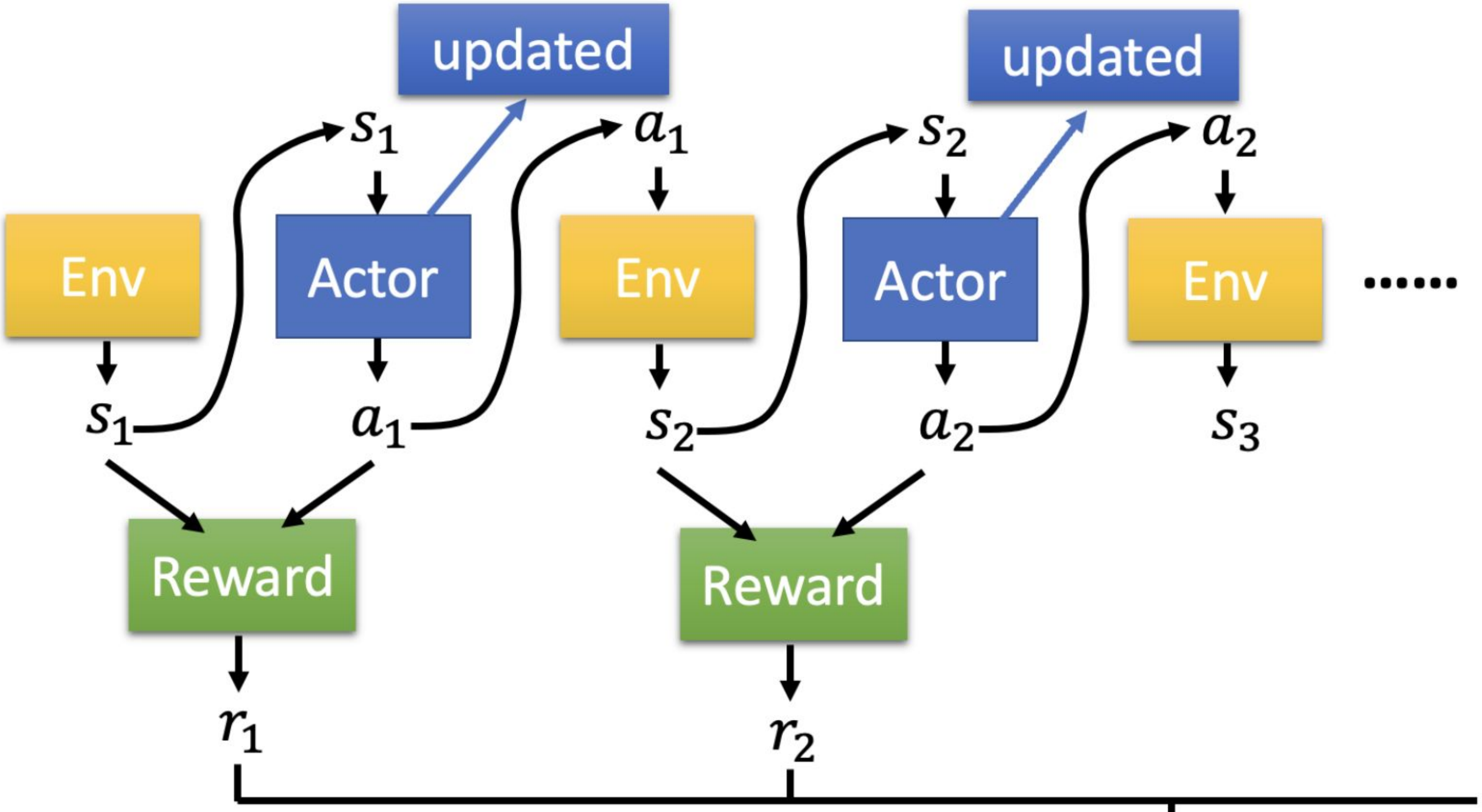


Trajectory $\tau = \{s_1, a_1, s_2, a_2, \cdots, s_T, a_T\}$

$$
\begin{aligned}
&p_\theta(\tau) \\
&= p(s_1)p_\theta(a_1|s_1)p(s_2|s_1,a_1)p_\theta(a_2|s_2)p(s_3|s_2,a_2)\cdots \\
&= p(s_1)\prod_{t=1}^{T} p_\theta(a_t|s_t)p(s_{t+1}|s_t,a_t)
\end{aligned}
$$

https://blog.csdn.net/qq_30615903

Reinforcement Learning

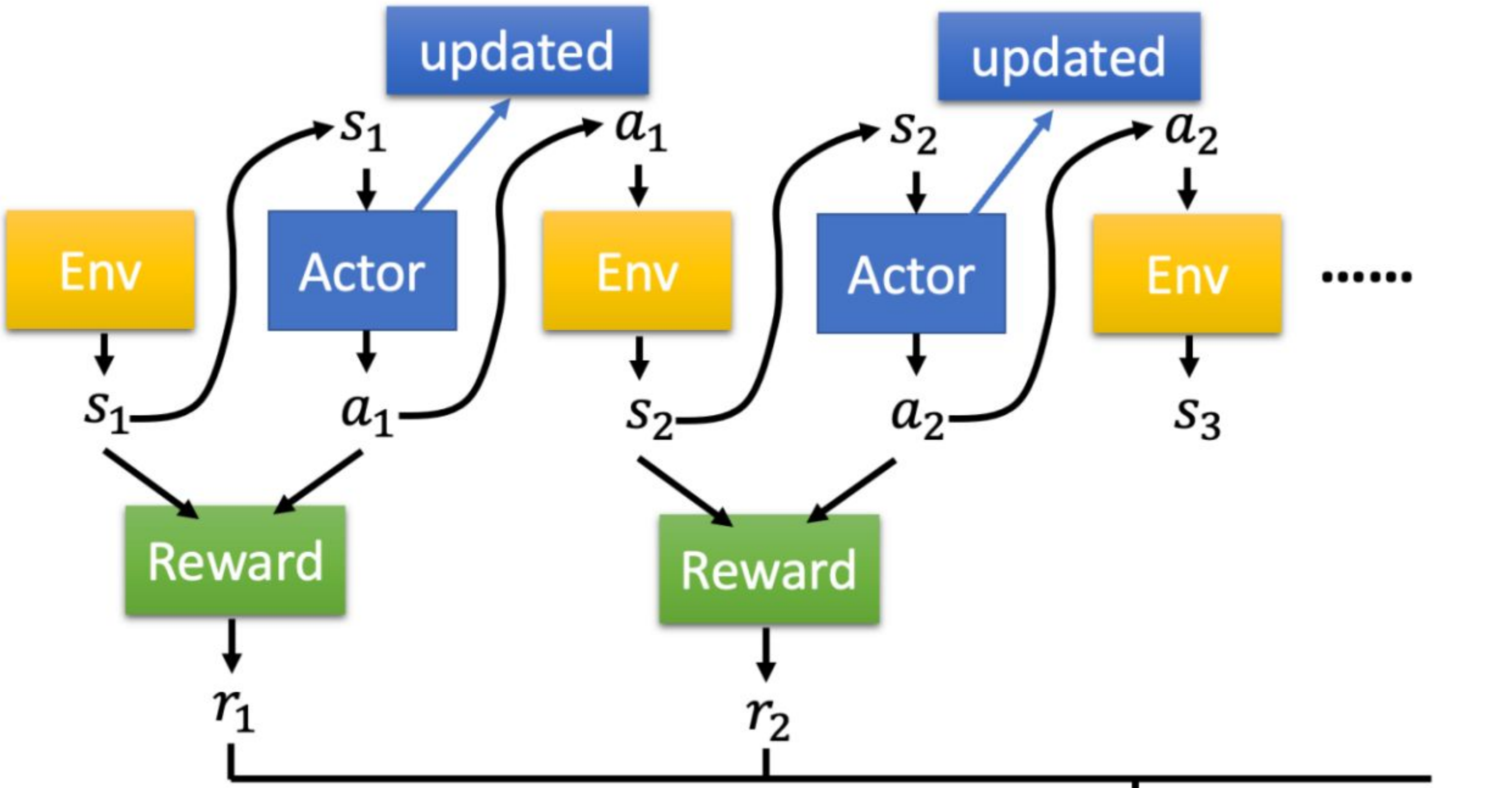


Expected Reward

$$\bar{R}_\theta = \sum_{\tau} R(\tau) p_\theta(\tau) = E_{\tau \sim p_\theta(\tau)}[R(\tau)]$$

$$R(\tau) = \sum_{t=1}^{T} r_t$$

Reinforcement Learning



Expected Reward

$$\bar{R}_\theta = \sum_\tau R(\tau)p_\theta(\tau) = E_{\tau\sim p_\theta(\tau)}[R(\tau)]$$

$$R(\tau) = \sum_{t=1}^{T} r_t$$

Solutions

- Q-Learning
- Policy Gradient
- Actor-Critic
- ...

Optimizing for Human Preferences

How do we change the LM parameters θ to maximize this?

$$\mathbb{E}_{\hat{s}\sim p_\theta(s)}[R(\hat{s})]$$

Gradient Ascent

$$\theta_{t+1} \coloneqq \theta_t + \alpha \nabla_{\theta_t} \mathbb{E}_{\hat{s}\sim p_{\theta_t}(s)}[R(\hat{s})]$$

Policy Gradient Methods in Reinforcement Learning (REINFORCE) [Williams, 1992]

Policy Gradient/REINFORCE

Gradient Ascent

$$\theta_{t+1} \coloneqq \theta_t + \alpha \nabla_{\theta_t} \mathbb{E}_{\hat{s} \sim p_{\theta_t}(s)}[R(\hat{s})]$$

$$\nabla_\theta \mathbb{E}_{\hat{s} \sim p_\theta(s)}[R(\hat{s})] = \nabla_\theta \sum_s R(s) p_\theta(s) = \sum_s R(s)\, \nabla_\theta p_\theta(s)$$

Log-Derivative Trick

$$\nabla_\theta \log p_\theta(s) = \frac{1}{p_\theta(s)} \nabla_\theta p_\theta(s) \quad \Longrightarrow \quad \nabla_\theta p_\theta(s) = \nabla_\theta \log p_\theta(s)\, p_\theta(s)$$

Policy Gradient/REINFORCE

$$\nabla_\theta \mathbb{E}_{\hat{s}\sim p_\theta(s)}[R(\hat{s})] = \sum_s R(s)\, \nabla_\theta p_\theta(s) = \sum_s p_\theta(s) R(s)\, \nabla_\theta \log p_\theta(s)$$

$$= \mathbb{E}_{\hat{s}\sim p_\theta(s)}[R(\hat{s})\, \nabla_\theta \log p_\theta(\hat{s})]$$

We can approximate this objective with Monte Carlo samples

$$\nabla_\theta \mathbb{E}_{\hat{s}\sim p_\theta(s)}[R(\hat{s})] = \mathbb{E}_{\hat{s}\sim p_\theta(s)}[R(\hat{s})\, \nabla_\theta \log p_\theta(\hat{s})] \approx \frac{1}{m}\sum_{i=1}^{m} R(s_i)\, \nabla_\theta \log p_\theta(s_i)$$

Policy Gradient/REINFORCE

Take gradient steps to maximize $p_\theta(s_i)$

If R is +++

$$\theta_{t+1} := \theta_t + \alpha \frac{1}{m} \sum_{i=1}^{m} R(s_i)\, \nabla_{\theta_t} \log p_{\theta_t}(s_i)$$

If R is ---

Take steps to minimize $p_\theta(s_i)$

We reinforce good actions, increasing the chance they happen again

Proximal Policy Optimization (PPO)

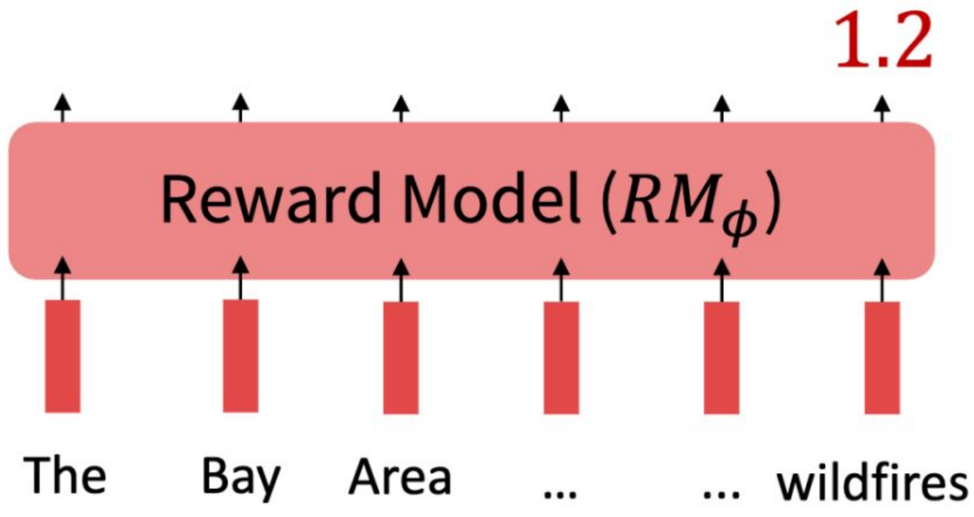
- New parameters θ' cannot be very different from old parameters θ

$$J_{PPO}^{\theta'}(\theta) = J^{\theta'}(\theta) - \beta KL(\theta, \theta')$$

Regularization

How to Model Human Preferences?

- Now for any reward function R, we can train our language model to maximize expected reward
- Problem 1: human-in-the-loop is expensive
  - Solution: instead of directly asking humans for preferences, model their preferences as a separate (NLP) problem
  - Train a reward model (RM) from an annotated dataset

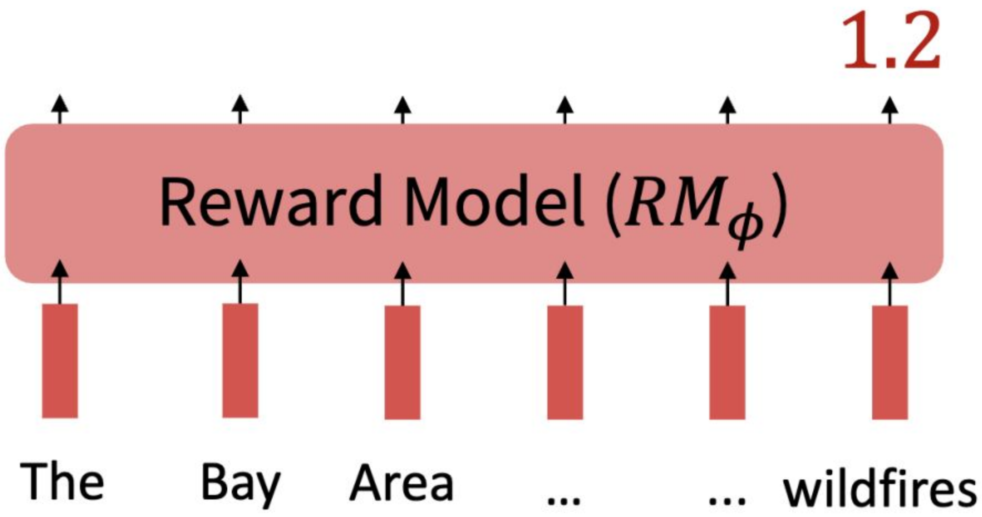


How to Model Human Preferences?

- Now for any reward function R, we can train our language model to maximize expected reward
- Problem 2: human judgments are noisy and miscalibrated
  - Solution: instead of asking for direct ratings, ask for pairwise comparisons, which can be more reliable

`An earthquake hit San Francisco. There was minor property damage, but no injuries.`	>	`A 4.2 magnitude earthquake hit San Francisco, resulting in massive damage.`	>	`The Bay Area has good weather but is prone to earthquakes and wildfires.`
s_1		s_3		s_2

Training A Reward Model



Bradley-Terry [1952] paired comparison model

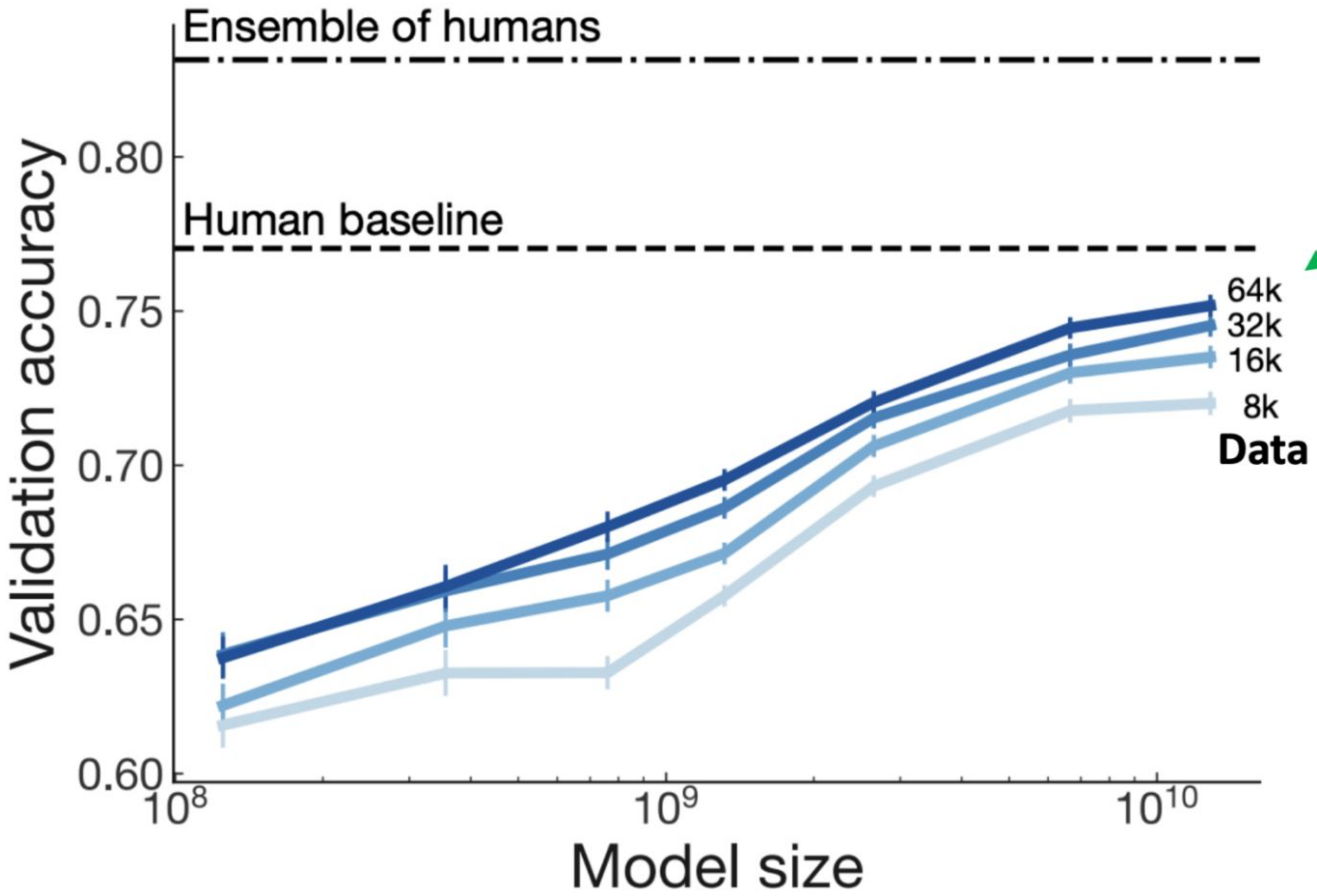
$$J_{RM}(\phi) = -\mathbb{E}_{(s^w, s^l)\sim D}\left[\log \sigma(RM_\phi(s^w) - RM_\phi(s^l))\right]$$

"winning" sample

"losing" sample

s^w should score higher than s^l

Reward Model vs. Real Human Feedback



Large enough RM trained on enough data approaching single human perf

[Stiennon et al., 2020]

RLHF: Putting Everything All Together

- We have the following:
  - A pretrained (possibly instruction-finetuned) LM $p^{PT}(y \mid x)$
  - A reward model $RM_\phi(x, y)$ that produces scalar rewards for LM outputs, trained on a dataset of human comparisons
- Now to do RLHF:
  - Copy the model $p_\theta^{RL}(y \mid x)$, with parameters θ we would like to optimize
  - We want to optimize:

$$\mathbb{E}_{\hat{y} \sim p_\theta^{RL}(\hat{y}|x)} \left[RM_\phi(x, \hat{y})\right]$$

RLHF: Putting Everything All Together

- We want to optimize:

$$\mathbb{E}_{\hat{y} \sim p_\theta^{RL}(\hat{y}|x)} \left[RM_\phi(x, \hat{y}) \right]$$

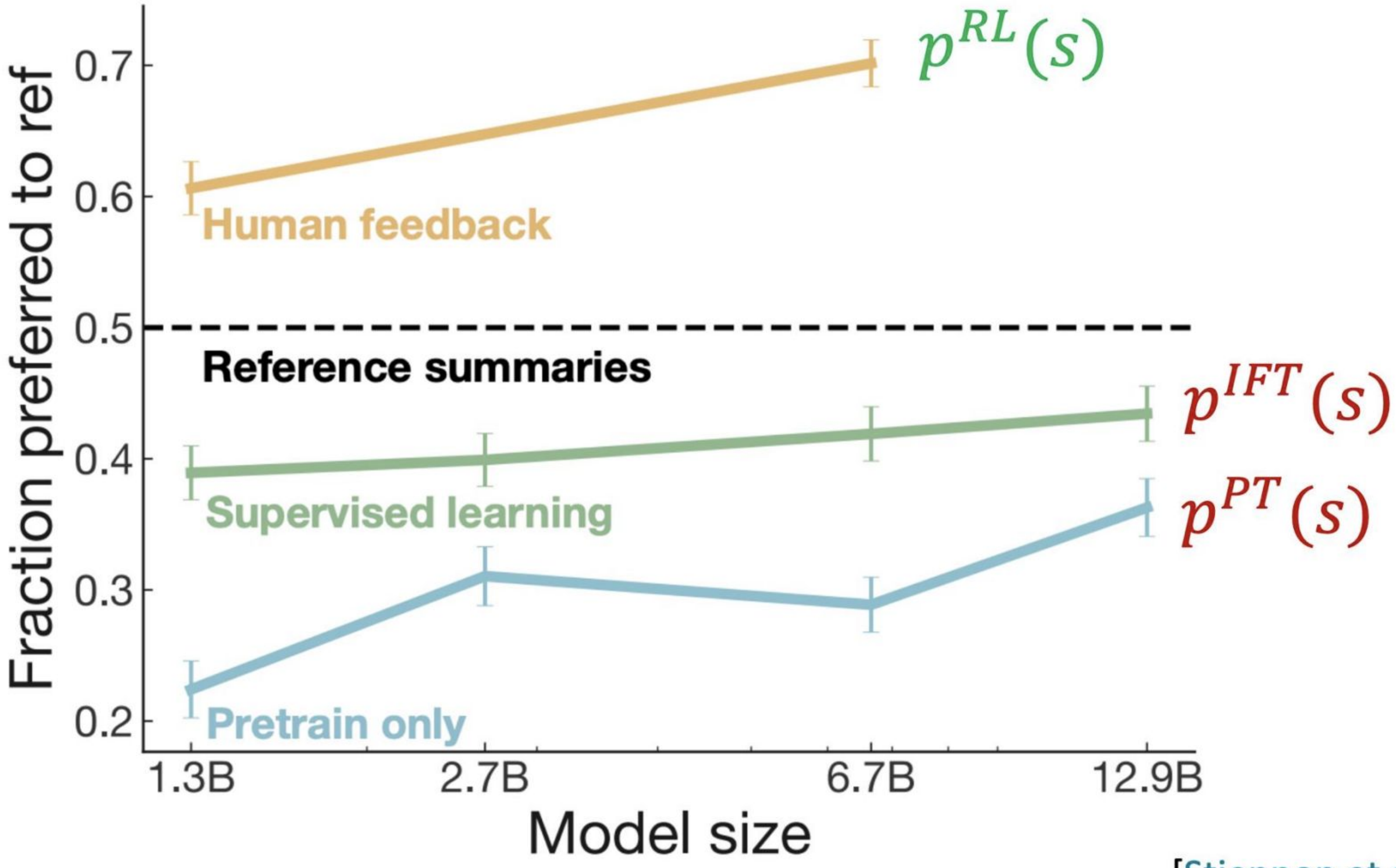
- Do you see any problems?
  - Learned rewards are imperfect; this quantity can be imperfectly optimized
- Add a penalty for drifting too for from the initialization:

$$\mathbb{E}_{\hat{y} \sim p_\theta^{RL}(\hat{y}|x)} \left[RM_\phi(x, \hat{y}) \; - \beta \underbrace{\log \left(\frac{p_\theta^{RL}(\hat{y} \mid x)}{p^{PT}(\hat{y} \mid x)} \right)} \right]$$

Pay a price when
$p_\theta^{RL}(\hat{y} \mid x) > p^{PT}(\hat{y} \mid x)$

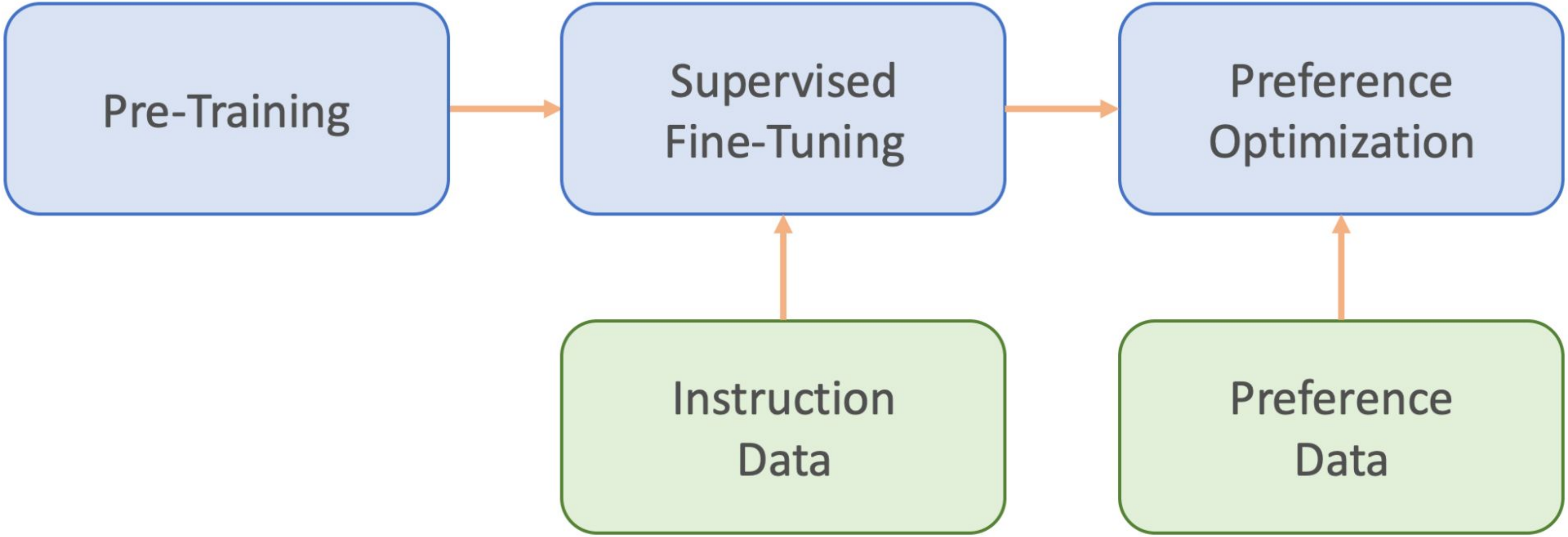
This penalty which prevents us from diverging too far from the pretrained model. In expectation, it is known as the **Kullback-Leibler (KL)** divergence between $p_\theta^{RL}(\hat{y} \mid x)$ and $p^{PT}(\hat{y} \mid x)$.

RLHF vs. SFT

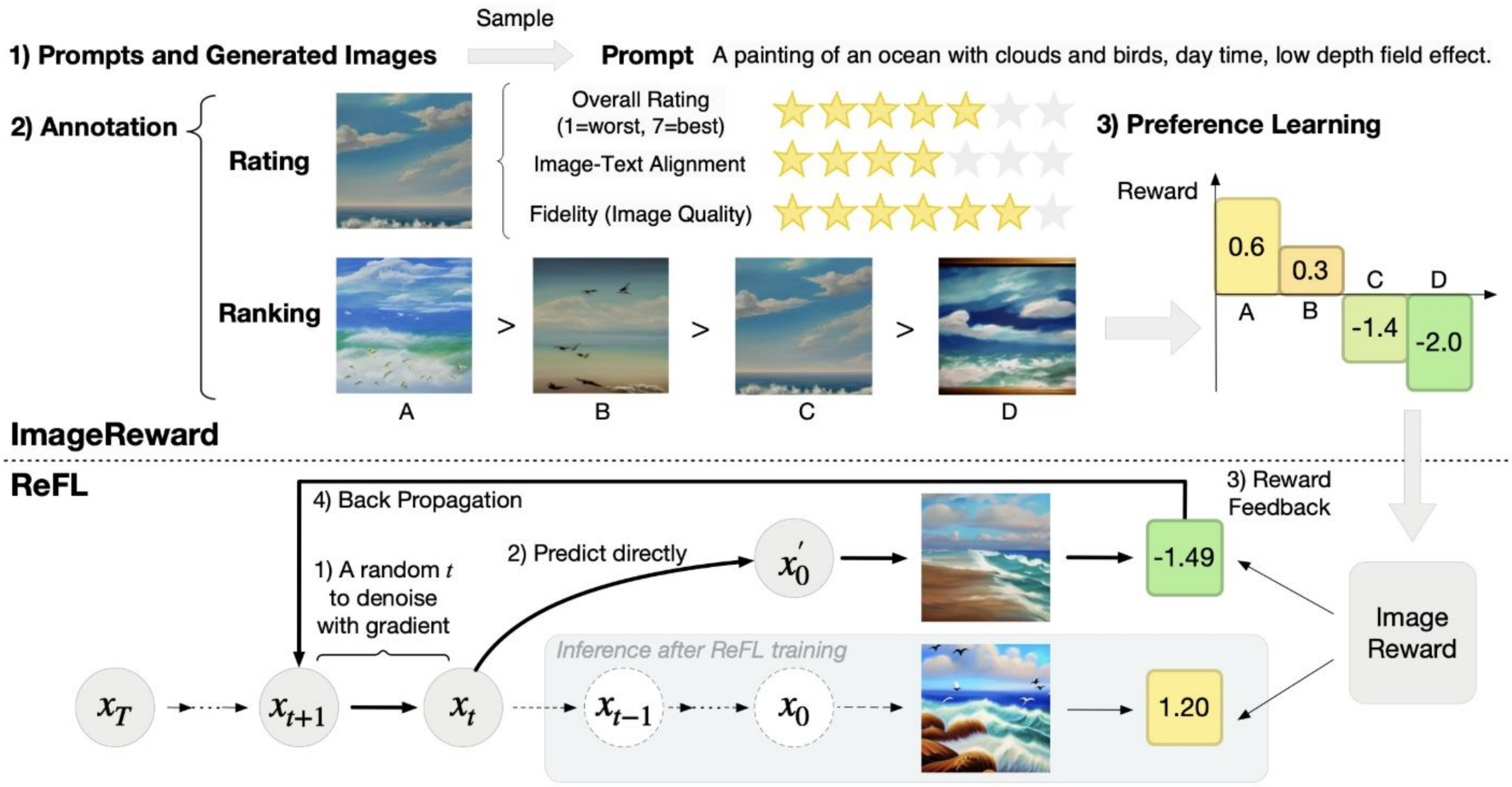


[Stiennon et al., 2020]

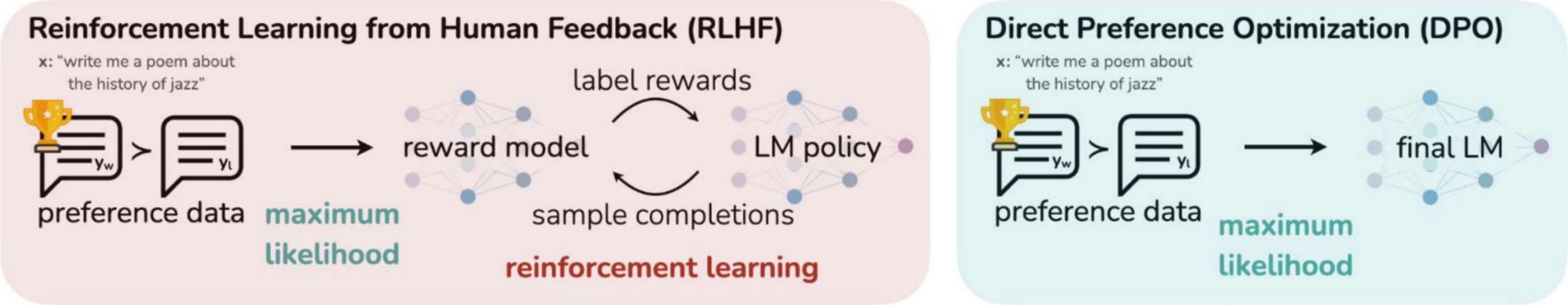
Alignment Pipeline



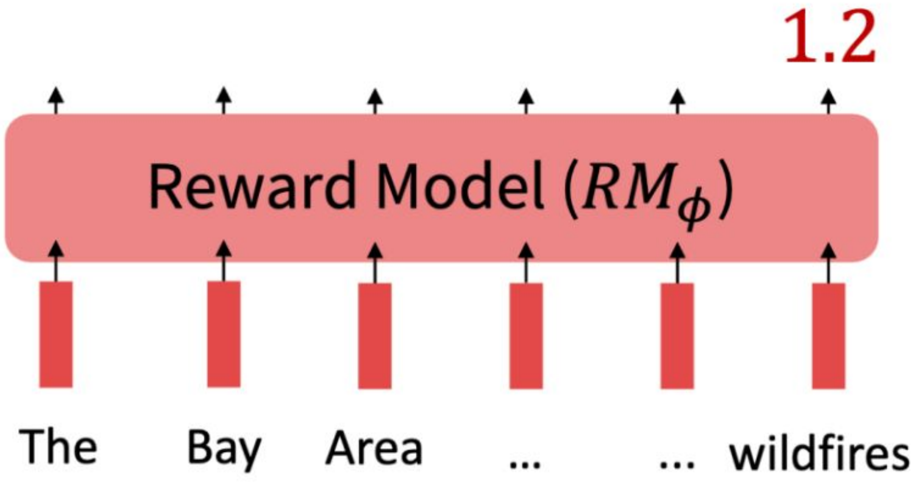
Extend to Text-to-Image Generation



Direct Preference Optimization (DPO)



RLHF: Proximal Policy Optimization (PPO)



An earthquake hit San Francisco. There was minor property damage, but no injuries.

s_1

>

The Bay Area has good weather but is prone to earthquakes and wildfires.

s_2

$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l)) \right]$$

Direct Preference Optimization (DPO)

RLHF Objective

(get **high reward**, stay **close** to reference model)

$$\max_{\pi} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi(y|x)} \left[r(x,y)\right] - \beta \mathbb{D}_{\mathrm{KL}}(\pi(\cdot \mid x) \| \pi_{\mathrm{ref}}(\cdot \mid x))$$

Maximize reward

Keep similar behavior

$$\max_{\pi} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi} \left[r(x,y)\right] - \beta \mathbb{D}_{\mathrm{KL}}\left[\pi(y|x) \mid\mid \pi_{\mathrm{ref}}(y|x)\right]$$

$$= \max_{\pi} \mathbb{E}_{x\sim\mathcal{D}} \mathbb{E}_{y\sim\pi(y|x)} \left[r(x,y) - \beta \log \frac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)}\right]$$

$$= \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}} \mathbb{E}_{y\sim\pi(y|x)} \left[\log \frac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)} - \frac{1}{\beta} r(x,y)\right]$$

$$= \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}} \mathbb{E}_{y\sim\pi(y|x)} \left[\log \frac{\pi(y|x)}{\frac{1}{Z(x)} \pi_{\mathrm{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x,y)\right)} - \log Z(x)\right]$$

$$Z(x) = \sum_{y} \pi_{\mathrm{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x,y)\right)$$

Direct Preference Optimization (DPO)

RLHF Objective

(get **high reward**, stay **close** to reference model)

$$\max_{\pi} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi(y|x)} \left[r(x,y)\right] - \beta \mathbb{D}_{\mathrm{KL}}(\pi(\cdot \mid x) \| \pi_{\mathrm{ref}}(\cdot \mid x))$$

Maximize reward

Keep similar behavior

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\mathrm{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x,y)\right)$$

$$\min_{\pi} \mathbb{E}_{x\sim\mathcal{D}} \mathbb{E}_{y\sim\pi(y|x)} \left[\log \frac{\pi(y|x)}{\frac{1}{Z(x)} \pi_{\mathrm{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x,y)\right)} - \log Z(x)\right]$$

$$= \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}} \left[\mathbb{E}_{y\sim\pi(y|x)} \left[\log \frac{\pi(y|x)}{\pi^*(y|x)}\right] - \log Z(x)\right]$$

$$= \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}} \left[\mathbb{D}_{\mathrm{KL}}(\pi(y|x) \mid\mid \pi^*(y|x)) - \log Z(x)\right]$$

$$\pi(y|x) = \pi^*(y|x) = \frac{1}{Z(x)} \pi_{\mathrm{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x,y)\right)$$

Direct Preference Optimization (DPO)

RLHF Objective

(get **high reward**, stay **close** to reference model)

$$\max_{\pi} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi(y|x)} [r(x,y)] - \beta \mathbb{D}_{\mathrm{KL}}(\pi(\cdot \mid x) \| \pi_{\mathrm{ref}}(\cdot \mid x))$$

Maximize reward

Keep similar behavior

Closed-form Optimal Policy

(write **optimal policy** as function of **reward function**; from prior work)

$$\pi^*(y \mid x) = \frac{1}{Z(x)} \pi_{\mathrm{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x,y)\right)$$

with $Z(x) = \sum_y \pi_{\mathrm{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x,y)\right)$

Note **intractable sum** over possible responses; can't immediately use this

Rearrange

(write **any reward function** as function of **optimal policy**)

Ratio is **positive** if policy likes response more than reference model, **negative** if policy likes response less than ref. model

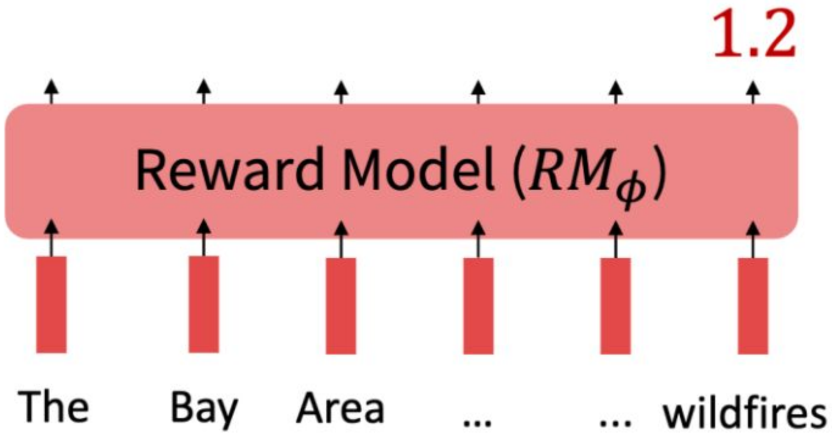
$$r(x,y) = \underbrace{\beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)}_{\text{some parameterization of a reward function}}$$

Direct Preference Optimization (DPO)

Derived from the Bradley-Terry model of human preferences:

A loss function on reward functions

$$\mathcal{L}_R(r, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma(r(x, y_w) - r(x, y_l)) \right]$$




```
An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.
```

$>$

```
The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.
```

s_1 s_2

Direct Preference Optimization (DPO)

Derived from the Bradley-Terry model of human preferences:

A loss function on reward functions

$$\mathcal{L}_R(r, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma(r(x, y_w) - r(x, y_l)) \right]$$

+

A transformation between reward functions and policies

$$r_{\pi_\theta}(x, y) = \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$$

Direct Preference Optimization (DPO)

Derived from the Bradley-Terry model of human preferences:

A loss function on reward functions

$$\mathcal{L}_R(r, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma(r(x, y_w) - r(x, y_l)) \right]$$

+

A transformation between reward functions and policies

$$r_{\pi_\theta}(x, y) = \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$$

=

Reward of **preferred** response

Reward of **dispreferred** response

A loss function on policies

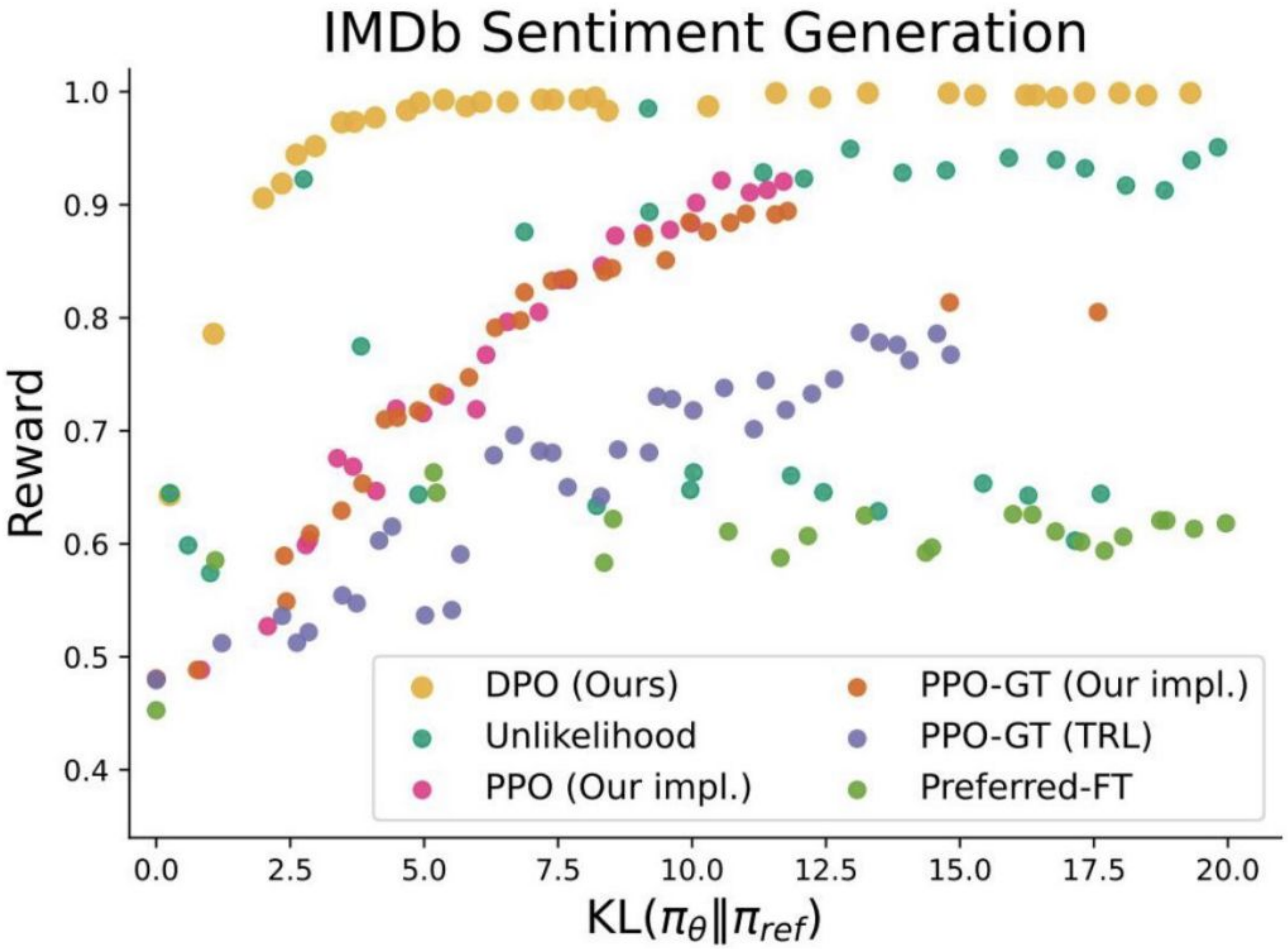
$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]$$

Direct Preference Optimization (DPO)

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\underbrace{\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)}}_{\text{Reward of preferred response}} - \underbrace{\beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}}_{\text{Reward of dispreferred response}}\right)\right]$$

$$\nabla_\theta\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\beta\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\underbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}_{\text{higher weight when reward estimate is wrong}}\left[\underbrace{\nabla_\theta\log\pi(y_w \mid x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_\theta\log\pi(y_l \mid x)}_{\text{decrease likelihood of } y_l}\right]\right]$$

DPO Performance

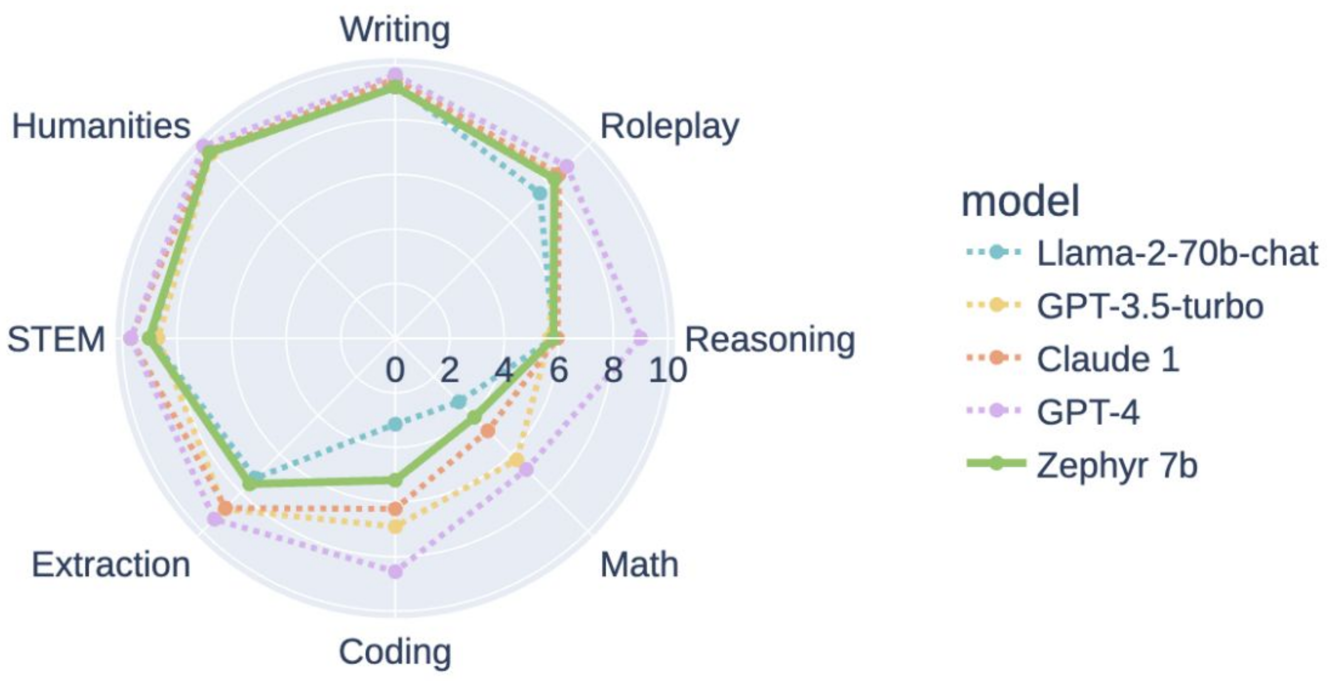


1. Generate positive IMDB reviews from GPT2-XL
2. Use pre-trained sentiment classifier as Gold RM
3. Create preferences based on Gold RM
4. Optimize with PPO and DPO

Large-Scale DPO Training

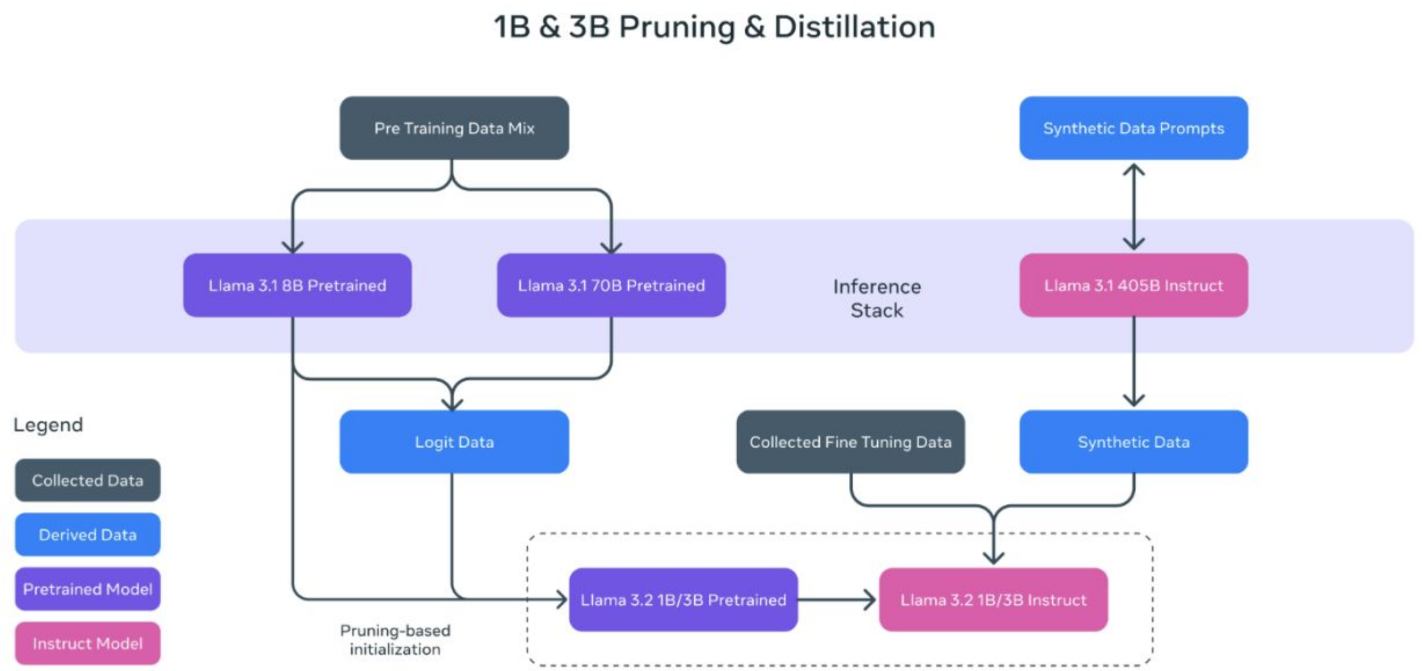
ZEPHYR: DIRECT DISTILLATION OF LM ALIGNMENT

Lewis Tunstall,* Edward Beeching,* Nathan Lambert, Nazneen Rajani, Kashif Rasul, Younes Belkada, Shengyi Huang, Leandro von Werra, Clémentine Fourrier, Nathan Habib, Nathan Sarrazin, Omar Sanseviero, Alexander M. Rush, and Thomas Wolf
The H4 (Helpful, Honest, Harmless, Huggy) Team
https://huggingface.co/HuggingFaceH4
lewis@huggingface.co



Large-Scale DPO Training

Llama 3.2: Revolutionizing edge AI and vision with open, customizable models



In post-training, we use a similar recipe as Llama 3.1 and produce final chat models by doing several rounds of alignment on top of the pre-trained model. Each round involves supervised fine-tuning (SFT), rejection sampling (RS), and direct preference optimization (DPO).

Statistical Rejection Sampling Optimization (RSO)

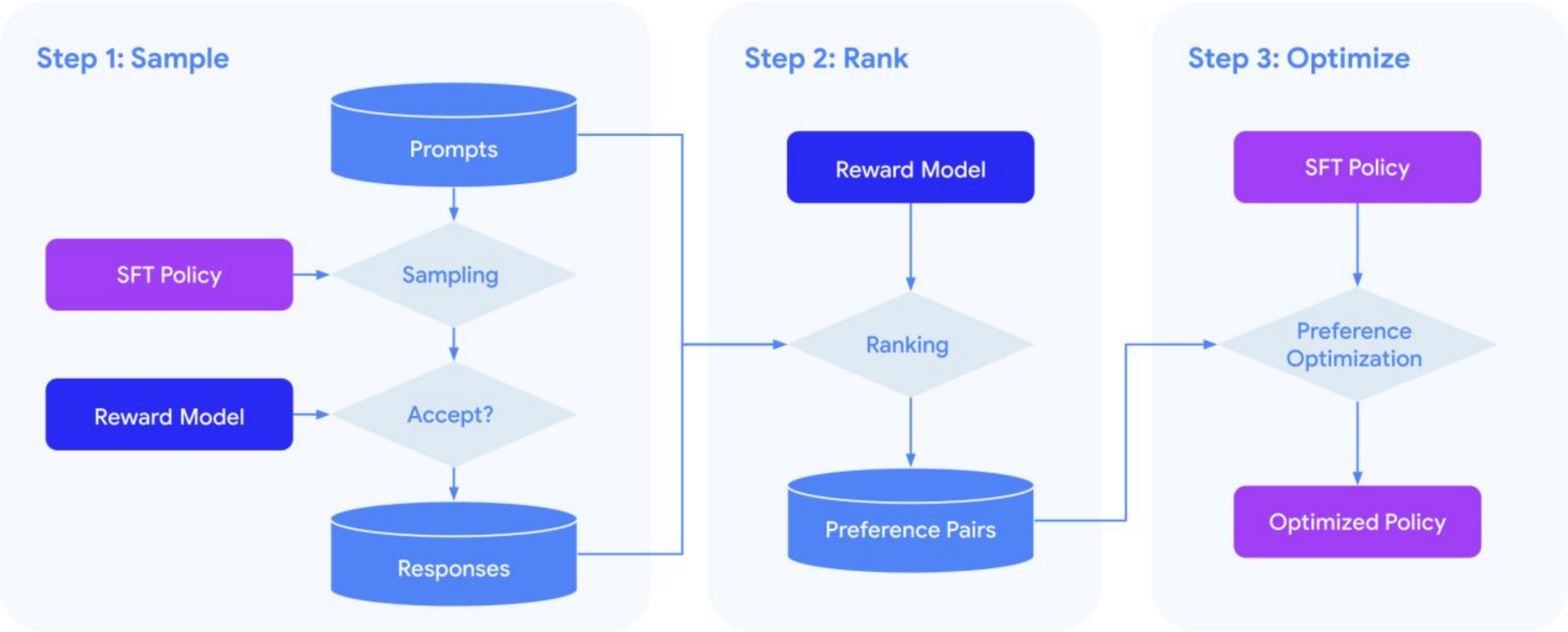


Figure 1: RSO first fits a pairwise reward-ranking model from human preference data. This model is later applied to generate preference pairs with candidates sampled from the optimal policy, followed by a preference optimization step to align sequence likelihood towards preferences.

Statistical Rejection Sampling Optimization (RSO)

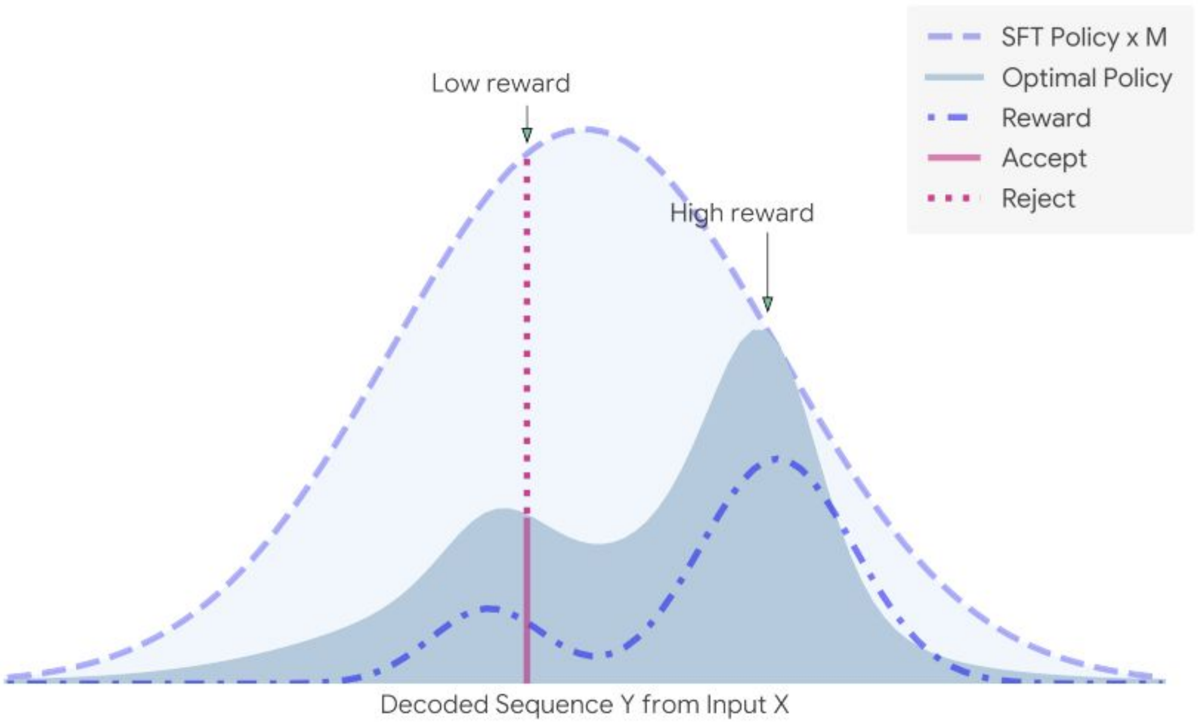


Figure 2: Statistical rejection sampling illustration. There are three curves in the figure: M times SFT policy, reward, optimal policy. The sample is first generated by SFT policy, then gets accepted or rejected depending on whether a uniform random variable locates in acceptance or rejection region. If the sample has high SFT policy probability but low optimal policy probability and reward score, it has a higher chance of being rejected.

Statistical Rejection Sampling Optimization (RSO)

Approach	Ablation		Metrics		
	Loss	Preference Pair	Proxy Reward (%)	Gold Reward (%)	AutoSxS (%)
		Reddit TL;DR			
RAFT	cross-entropy	-	74.84	68.51	53.77
ReST	cross-entropy	-	49.03	46.17	34.36
DPO	sigmoid-norm	direct	84.35	76.09	67.72
	sigmoid-norm	sft-sample-rank	88.63	78.14	69.02
$\text{RSO}_{\text{sigmoid-norm}}$	sigmoid-norm	rso-sample-rank	**92.37**	**82.22**	**71.86**
$\text{SLiC}_{\text{direct}}$	hinge	direct	86.92	79.76	60.54
$\text{SLiC}_{\text{sample-rank}}$	hinge	sft-sample-rank	90.15	80.19	67.34
	hinge	rso-sample-rank	**93.36**	**84.40**	**69.26**
	hinge-norm	direct	83.93	76.43	66.63
	hinge-norm	sft-sample-rank	88.04	76.57	68.46
$\text{RSO}_{\text{hinge-norm}}$	hinge-norm	rso-sample-rank	**92.80**	**83.45**	**70.84**
		AnthropicHH			
RAFT	cross-entropy	-	58.21	40.00	24.99
ReST	cross-entropy	-	43.48	30.33	15.58
DPO	sigmoid-norm	direct	51.63	36.13	24.01
	sigmoid-norm	sft-sample-rank	85.09	58.65	39.56
$\text{RSO}_{\text{sigmoid-norm}}$	sigmoid-norm	rso-sample-rank	**86.94**	**59.15**	**40.98**
$\text{SLiC}_{\text{direct}}$	hinge	direct	35.95	27.56	15.69
$\text{SLiC}_{\text{sample-rank}}$	hinge	sft-sample-rank	80.82	54.55	30.66
	hinge	rso-sample-rank	**82.21**	**55.22**	**32.56**
	hinge-norm	direct	49.55	37.23	22.89
	hinge-norm	sft-sample-rank	82.40	56.55	35.96
$\text{RSO}_{\text{hinge-norm}}$	hinge-norm	rso-sample-rank	**84.44**	**57.75**	**38.58**

Table 1: Compare different methods with T5-large policy to leverage human feedback data. Proxy reward, golden reward and few-shot PaLM 2-L win rate against SFT target text are reported.

RSO - LLama 2-Chat

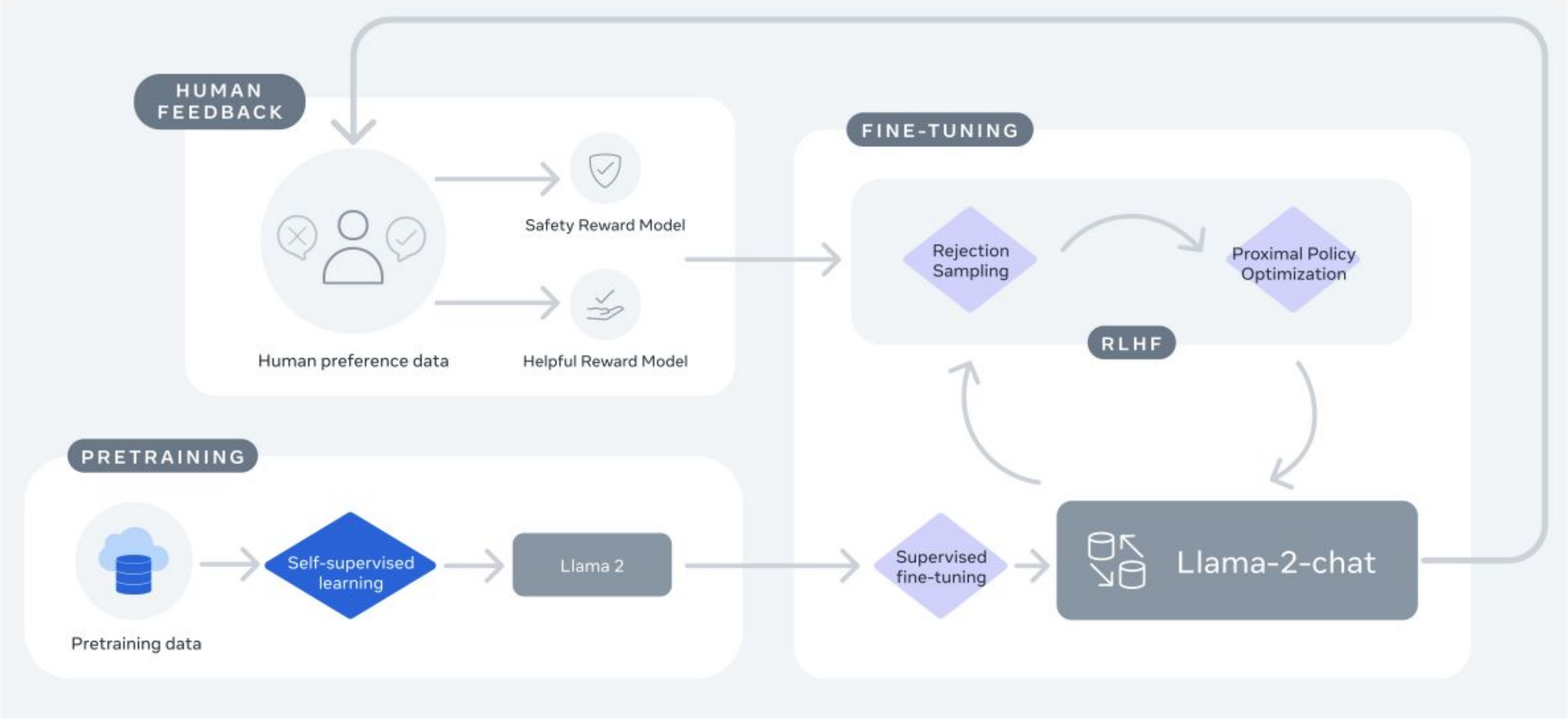
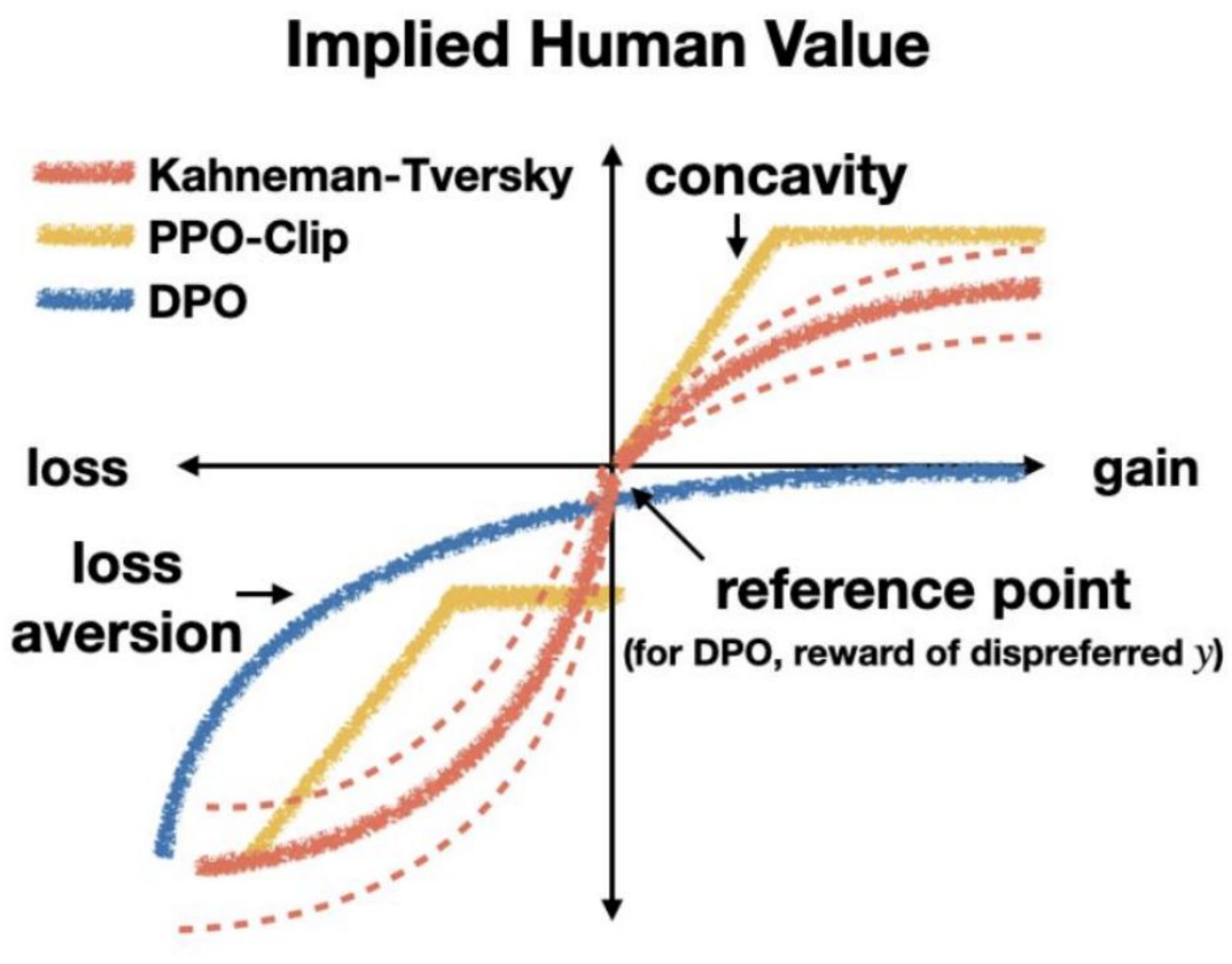


Figure 4: Training of Llama 2-Chat: This process begins with the **pretraining** of Llama 2 using publicly available online sources. Following this, we create an initial version of Llama 2-Chat through the application of **supervised fine-tuning**. Subsequently, the model is iteratively refined using Reinforcement Learning with Human Feedback (**RLHF**) methodologies, specifically through rejection sampling and Proximal Policy Optimization (PPO). Throughout the RLHF stage, the accumulation of **iterative reward modeling data** in parallel with model enhancements is crucial to ensure the reward models remain within distribution.

Kahneman-Tversky Optimization (KTO)



Which One Do You Choose?

- Imagine you are facing two choices:
  - **Choice one**: has an 80% chance of earning you 10 million US dollars, and a 20% chance of giving you nothing
  - **Choice two**: gives you 4 million US dollars for sure

Which One Do You Choose?

- Imagine you are facing two choices:
  - **Choice one**: has an 80% chance of earning you 1 thousand US dollars, and a 20% chance of giving you nothing
  - **Choice two**: gives you 4 hundred US dollars for sure

Which One Do You Choose?

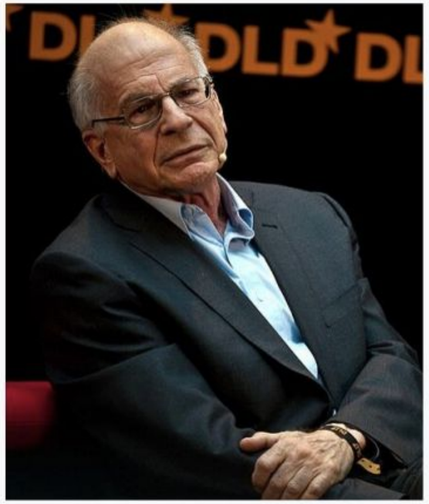
- Imagine you are facing two choices:
  - **Choice one**: has an 80% chance of earning you 10 US dollars, and a 20% chance of giving you nothing
  - **Choice two**: gives you 4 US dollars for sure

Prospect Theory

Prospect theory explains why humans make decisions about uncertain events that do not maximize expected value. It formalizes how humans perceive random variables in a biased but well-defined manner;
for example, relative to some **reference point**, humans are more sensitive to losses than gains, a property called **loss aversion**.

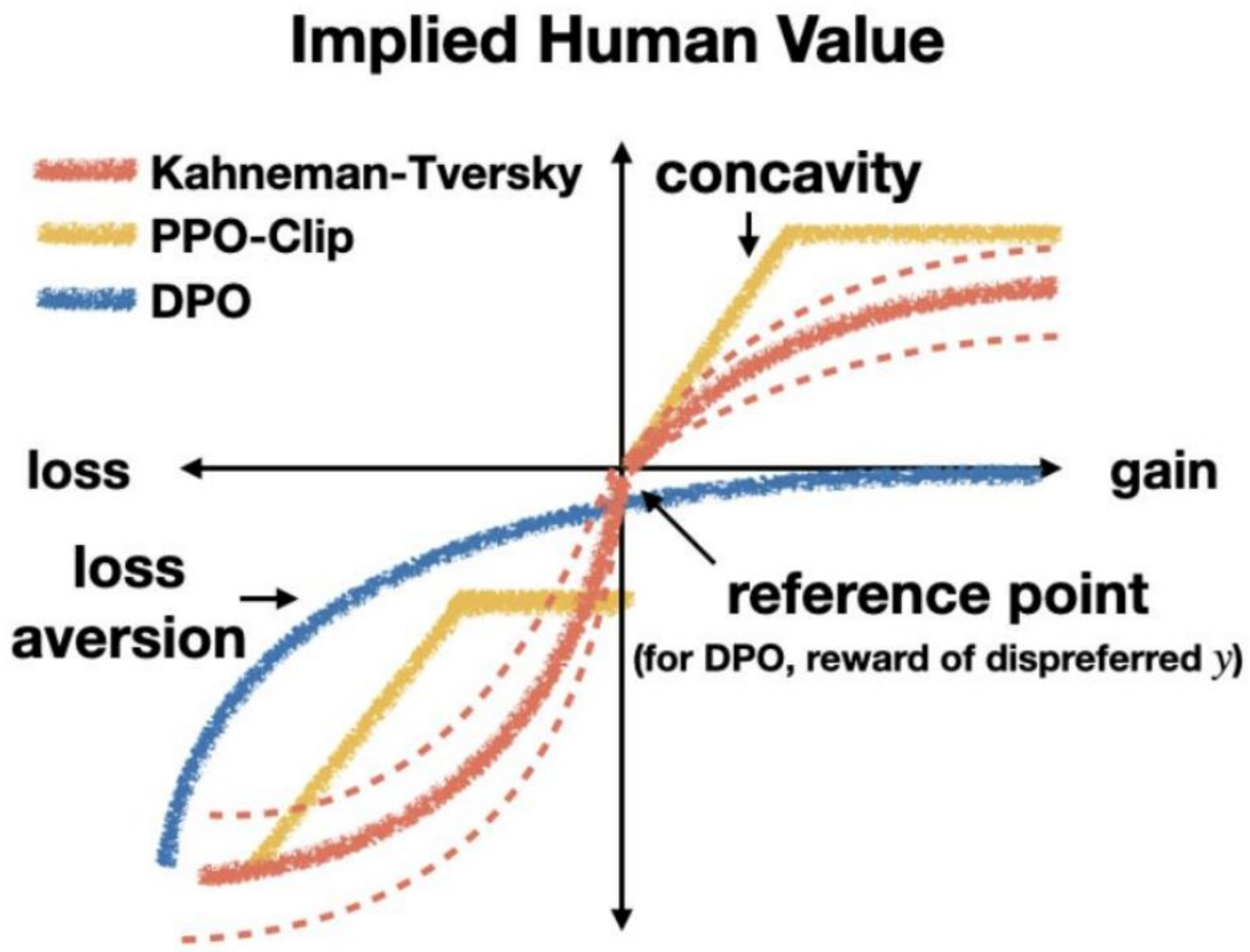
2002 Nobel Prize-winning economists

Daniel Kahneman

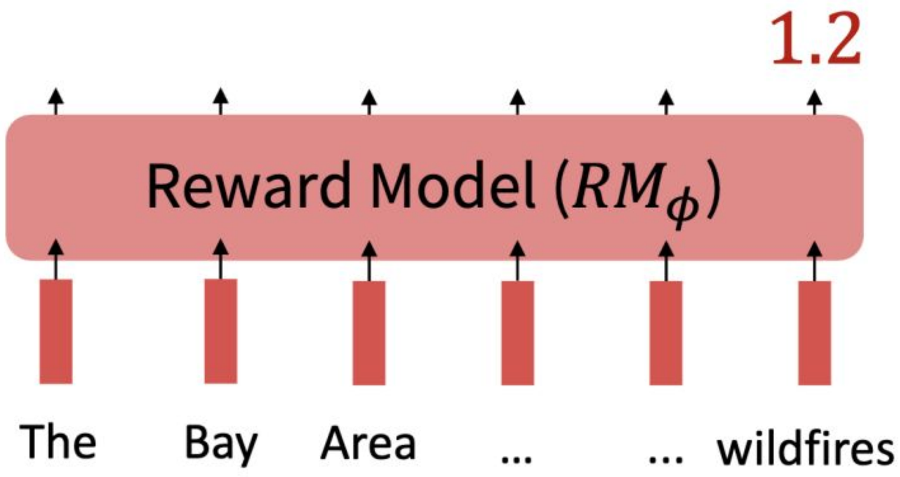


Amos Tversky

KTO Value Function



Preference Data For PPO/DPO



An earthquake hit San Francisco. There was minor property damage, but no injuries.

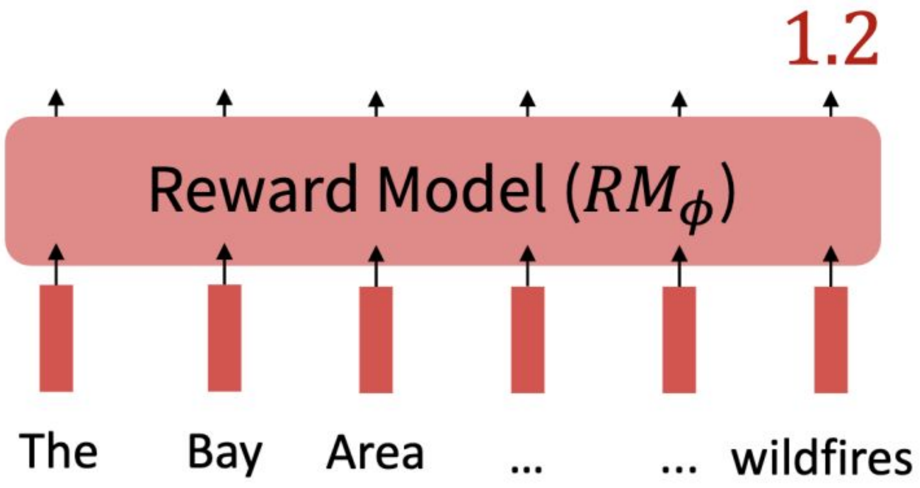
$>$

The Bay Area has good weather but is prone to earthquakes and wildfires.

s_1 s_2

Training Data (x, y_1, y_2)

Preference Data For KTO



An earthquake hit San Francisco. There was minor property damage, but no injuries.

s_1

Acceptable?

Training Data (x, y)

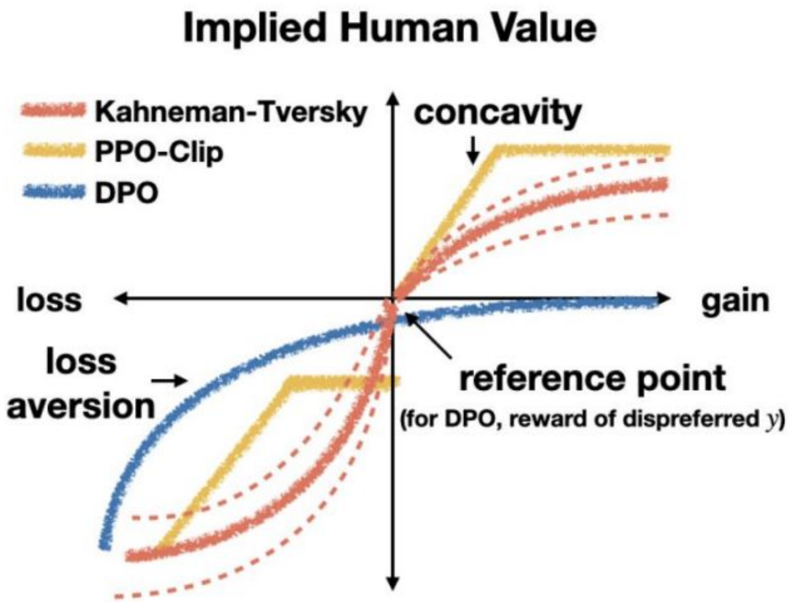
KTO: Loss Function

$$L_{\text{KTO}}(\pi_\theta, \pi_{\text{ref}}) = \mathbb{E}_{x,y\sim D}[\lambda_y - v(x,y)]$$

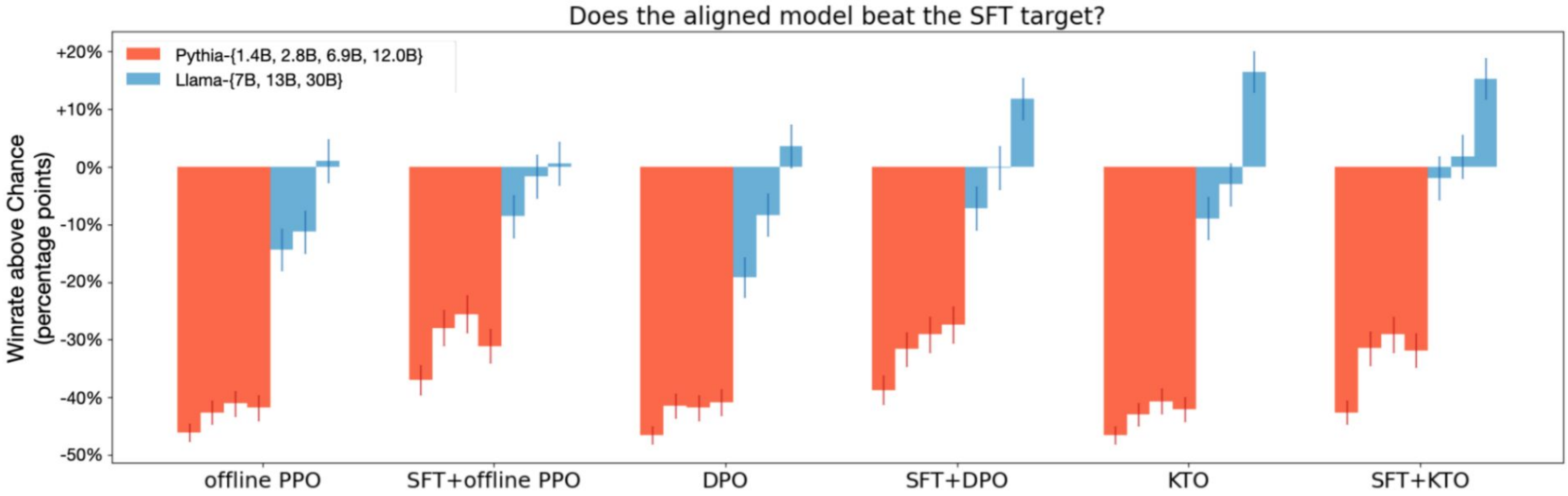
$$r_{\text{KTO}}(x,y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$$

$$v_{\text{KTO}}(x,y;\beta) = \begin{cases} \sigma(r_{\text{KTO}}(x,y) - z_{\text{ref}}) \text{ if } y \sim y_{\text{desirable}}|x \\ \sigma(z_{\text{ref}} - r_{\text{KTO}}(x,y)) \text{ if } y \sim y_{\text{undesirable}}|x \end{cases}$$

$$w(y) = \begin{cases} \lambda_D & \text{if } y \sim y_{\text{desirable}}|x \\ \lambda_U & \text{if } y \sim y_{\text{undesirable}}|x \end{cases}$$



KTO Performance



Thanks

Q&A

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